



UNIVERSITY OF
BIRMINGHAM



The PNSD Toolkit

User manual

James Brean

School of Geography, Earth and Environmental Sciences
University of Birmingham
j.brean@bham.ac.uk

Version 1.0 · September 2026

Contents

Contents.....	2
1. About PyNSD.....	3
1.1 What PyNSD can do.....	3
1.2 What PyNSD cannot do.....	4
2. Installing and starting	4
3. Your data	4
3.1 What a valid file looks like.....	4
3.2 Formats and layout	5
3.3 Column headers.....	5
3.4 Timestamps	5
3.5 Units	6
3.6 Files PyNSD cannot read.....	6
4. Load data	6
4.1 Adding files	7
4.2 Corrections	7
4.3 Combining files	8
5. Summary	8
6. Trend analysis.....	8
7. NPF identifier.....	9
8. Automated NPF identifiers.....	10
9. Cluster	11
10. Nano ranking	12
11. Wind.....	13
12. PMF	13
13. Pollution flags.....	15
14. What each panel exports.....	15
15. Limits and caveats.....	Error! Bookmark not defined.
16. References	Error! Bookmark not defined.
Appendix A. Error messages.....	16

1. About PyNSD

PyNSD started when I wanted to finally have a home-made tool to explore my Particle Number Size Distribution (PNSD) data and identify NPF events. As time went on I added additional modules, which brings us to the present software package.

PyNSD takes PNSDs and turns them into contour plots, diurnal cycles, growth rates, formation rates, clusters and source profiles. It runs as a desktop application, so you point it at a file and work through the panels on the left.

It was built for ambient measurements. Instrument exports carry preamble lines, odd delimiters, European decimal commas, repeated timestamps and gaps. PyNSD should hopefully handle all of these.

Every panel has a button to export the figures as bitmaps, and most have a button to export the working datasets.

PyNSD expects $dN/d\log D_p$ in cm^{-3} , and it will not warn you if you give it something else. Section 3 covers this.

1.1 What PyNSD can do

Area	What you get
Loading	Read CSV, text and Excel files with automatic delimiter, encoding, header and date detection. Merge, splice and harmonise files from different instruments.
Corrections	Spike removal, floor and ceiling limits, Gormley-Kennedy line loss, dN to $dN/d\log D_p$ normalisation, date and diameter filters. All reversible.
Visualisation	Contour plots, mean size distributions, diurnal cycles, total number time series, seasonal and day-of-week breakdowns.
NPF physics	Condensation sink, coagulation sink, growth rate, formation rate J , and $J_{1.5}$ by the Kerminen-Kulmala survival equation.
NPF identification	Day-by-day manual classification, a physical classifier, a trained CNN, and an automatic growth-rate tracker.
Statistics	Eight clustering algorithms with scoring and tuning, and nano-ranking after (Aliaga et al., 2023)
Source work	Wind sector analysis and a full PMF2 front-end with bootstrapping, tracer correlation and polar plots.

1.2 What PyNSD cannot do

It will not	Use instead
Invert raw mobility or optical data	Your instrument software. PyNSD starts from an inverted distribution.
Read instrument binary formats	Export to CSV first.
Run PMF on its own	PMF2 is licensed separately. PyNSD writes the input files, drives the executable and reads the output.
Classify days without the CNN weights	Download <code>NPF_CNN_model.pkl</code> separately. See section 2.
Propagate uncertainty	Nothing in PyNSD carries an error bar. Treat every number as a central estimate.
Handle more than one site at a time	Load one site, finish, then load the next.
Replace looking at your data	Every automatic classification is preliminary. Check the results!!!

2. Installing and starting

PyNSD is built and tested against Python 3.12. Install into a virtual environment so the dependencies stay separate from anything else on your machine.

- Clone or download the repository, then open a terminal in the `PyNSD` folder.
- Create the environment: `python -m venv venv`, then activate it with `venv\Scripts\activate` on Windows or `source venv/bin/activate` elsewhere.
- Install the dependencies: `pip install -r requirements.txt`.
- Start the application: `python main.py`.

The deep learning panel needs a separate file. Download `NPF_CNN_model.pkl` from the Code Availability section of Kecorius et al. (2024) and put it in `PyNSD/models/`. It is around 100 MB and is not included in the repository. Every other panel works without it.

PyNSD remembers what you had open. When you restart it, it offers to re-read the same files with the same import settings. It does not replay the corrections you applied, because applying them twice would double-count them. The panel lists them by name.

3. Your data

Most problems people have with PyNSD are problems with the file they gave it.

3.1 What a valid file looks like

One row per timestamp. One column for the time. One column per size bin, with the bin diameter written in the header. Values are $dN/d\log Dp$ in cm^{-3} .

datetime	10.6	11.5	12.4	...
2021-01-31 00:15:00	1240.5	1388.1	1502.7	...
2021-01-31 00:30:00	1198.2	1301.6	1477.3	...
2021-01-31 00:45:00	1302.9	1425.0	1610.4	...

Table 1. A file PyNSD will read without complaint. The header numbers are the geometric mean diameters of the bins, in nanometres.

3.2 Formats and layout

Feature	What is accepted
Extensions	.csv, .txt, .dat, .tsv, .xlsx, .xls, .xlsm
Delimiters	Comma, tab, semicolon, pipe or aligned whitespace. Detected by finding the field count that most lines agree on.
Encodings	UTF-8, Windows-1252 and Latin-1, tried in that order.
Preamble	Any number of instrument header lines above the real header row. They are skipped and the count is reported.
Decimals	Full stop or comma. A file written 1,234 for 1.234 is converted, and PyNSD says so.
Bin order	Ascending or descending. The columns are relabelled in file order, so a file running large to small is not silently reversed.
Bin spacing	Even or uneven. Every integral uses the real per-bin width, which matters for spliced datasets.

3.3 Column headers

A column becomes a size bin if its header carries a number that could be a diameter. Anything else is left alone.

Header	Read as	Why
10.6	10.6 nm	A bare number.
Dp_10.6nm	10.6 nm	Number with a unit, in a decorated label.
dN/dlogDp 10.6	10.6 nm	The number is found inside the label.
N(0.0106 um)	10.6 nm	Micrometres converted to nanometres.
1.06E-8	10.6 nm	No unit given, but the whole header row looks like metres.
Bin 12	Ignored	Index word. This is a channel number, not a size.
Channel 3	Ignored	Index word.
PM10, NO2, O3	Ignored	Species names, not diameters.
Unnamed: 4	Ignored	An empty column from a spreadsheet.
0.1	Ignored	Outside the plausible range of 0.3 nm to 100 μm .

Table 2. How PyNSD reads your column headers. Units are taken from the header where they are given. Where they are not, PyNSD infers nanometres, micrometres or metres from the size of the numbers and records the assumption in the file notes.

One more check runs at the ends. A numeric column sitting far off the log-spaced bin ladder, such as a column literally named 1, is dropped so it cannot pollute the distribution. This is skipped on datasets with ten bins or fewer, where losing a real bin would be worse.

3.4 Timestamps

Set the date column and format in **Load data**, or leave both on **Auto-detect**. Auto-detection scores every column by how well it parses and by its name.

Example in your file	How to handle it
2021-01-31 00:15:00	Detected, or set YYYY-MM-DD HH:mm:ss.
31/01/2021 00:15	Detected, or set DD/MM/YYYY HH:mm.
01/31/2021	Set MM/DD/YYYY explicitly if the day is never above 12.
44227.0104	Excel serial numbers are detected and converted.
Date in one column, time in another	Detected automatically and joined.
2021-01-31T00:15:00+00:00	Converted into the timezone you choose.
Anything else	Choose Custom... and type the format.

Where a date parses equally well two ways, PyNSD picks one and warns you. Check the file notes.

Out-of-order rows are sorted by datetime. Repeated timestamps are removed, keeping the first. Rows that fall in a daylight-saving gap are dropped.

3.5 Units

PyNSD assumes $dN/d\log Dp$ in cm^{-3} from the moment your file loads.

If your file holds dN instead, load it and use **Normalisation** in the **Load data** panel. PyNSD shows the mean logarithmic bin width and divides by it. **Un-normalise** reverses this. If your file holds raw counts per channel, convert them before you load.

3.6 Files PyNSD cannot read

The file	What happens	The fix
Bins as rows, times as columns	<i>No size-bin columns found</i>	Transpose it before loading.
Two header rows, size above unit	The header is misread	Flatten it to one header row.
Headers Bin 1 to Bin 30	<i>No size-bin columns found</i>	Rename the headers with the bin diameters.
No column that parses as a date	<i>No column in this file parses as a date</i>	Name the date column explicitly.
Raw counts per channel	Loads, and every result is wrong	Convert to $dN/d\log Dp$ first.
Fewer than three rows	Date detection needs at least three rows	Nothing to do. This is not a time series.

Table 3. The six ways a file fails, and what to do about each.

4. Load data

Add files, check how they were read, correct what needs correcting, then click **Proceed with selected dataset** to unlock the rest of the panels. The three numbered sections collapse, so you only see the step you are on.

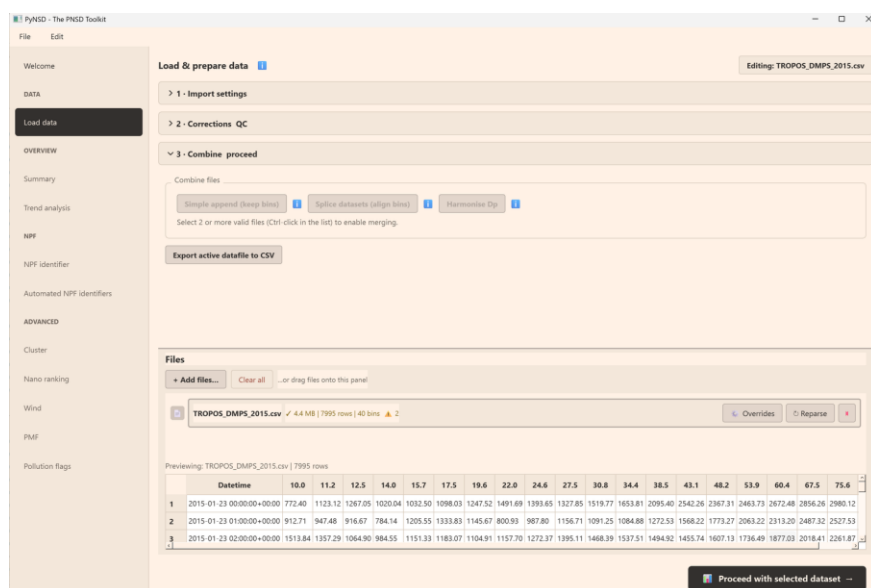


Figure 1. The **Load data** panel. Files appear at the bottom with row count, bin count and a status flag; the numbered sections above hold import settings, corrections and the combine tools. The warning triangle beside a file opens the notes.

4.1 Adding files

Click + **Add files...**, or drag files onto the panel. Each file appears in the list with its row count, bin count, date range and diameter range. A coloured status tells you whether it loaded cleanly, loaded with warnings, or failed.

Click the warning marker on any file to see everything PyNSD assumed and everything it dropped. Read this the first time you load a new instrument. It is the only place the unit assumption and the date-order warning appear.

Import settings apply globally, and you can override them per file. Use **Drop columns** to ignore columns by name, and **Error flag filtering** to delete rows where a flag column holds a given value.

4.2 Corrections

Every correction is reversible. **Undo** steps back one at a time, and the panel names what is on the stack, because applying normalise or line loss twice would compound the error.

Correction	What it does
QC spike removal	Flags points more than n standard deviations from a rolling median, plus all negative values. Replace them with the median or blank them. A diagnostic plot shows what moved.
Floor and ceiling	Replaces values below a threshold with the threshold. MPSS instruments cannot reliably measure below 1 cm^{-3} , and low values are sometimes error flags.
Line loss	Gormley-Kennedy diffusional losses from tube length, internal diameter, temperature and flow. The penetration curve is plotted before you apply it.
Normalisation	Converts between dN and dN/dlogDp using the bin widths.
Date and time filter	Keep a date range, or particular days of the week, months, hours or years. The plot shows what survives.
Diameter filter	Keep a sub-range of size bins.

4.3 Combining files

Select two or more files with Ctrl-click, then choose a mode.

- **Simple append** stacks files from the same instrument covering different periods. Bins missing from one file become blanks.
- **Splice** joins two instruments covering the same period with different bins, such as a nano-SMPS and a long-SMPS. You choose the crossover diameter and see the join before you accept it.
- **Harmonise** interpolates several datasets onto one common bin set using a monotonic CDF spline, which conserves total number and cannot produce negative concentrations.

5. Summary

Your first look at a dataset. The top panel is a contour plot of $dN/d\log D_p$ against time with total number overlaid. Zoom into any period and everything below follows.

The lower plots give the mean size and mass distributions with lognormal modes fitted, and the mean diurnal cycle of number and mass with its spread. The table on the right reports mean N, mean mass, the global mode and mean diameters, and the height and position of each fitted mode. Set the particle density in the box at the top (default 1.5 g cm^{-3}).

The PNSD in the bottom left will have 3-4 lognormal modes fit to it, the statistics for which are reported in the bottom right.

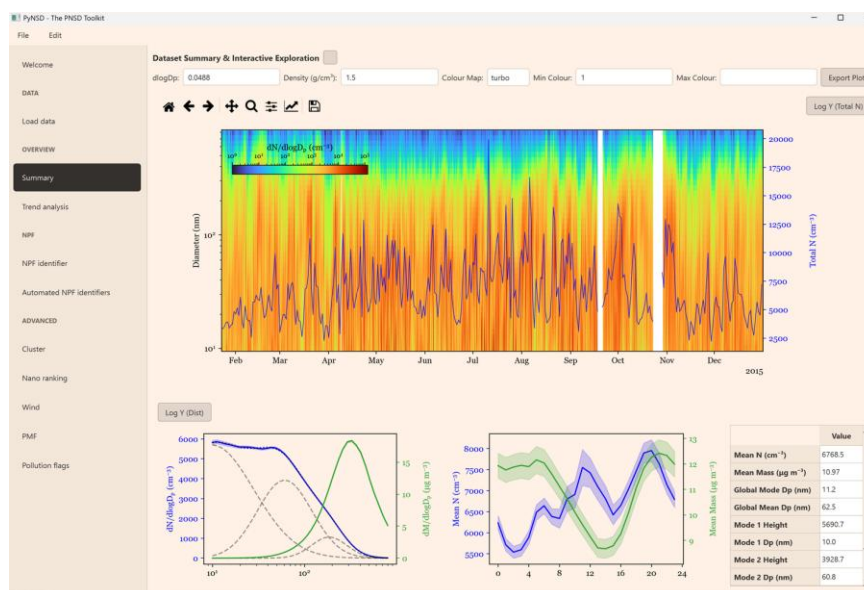


Figure 2. The **Summary** panel over a year of hourly data. Zooming the contour plot recomputes the distributions, the diurnal cycle and the statistics table beneath it.

6. Trend analysis

Long-term behaviour, with three models to choose between. Type the size fractions you want as ranges in nanometres, for example 10-30, 30-100, 100-1000, pick an averaging period, then run one of the models.

- **Linear regression** fits an ordinary least-squares line through number concentration, and through total mass on a second axis. You get the slope per year, R^2 and a p-value.

- **Mann-Kendall** tests for a monotonic trend without assuming linearity or normality. You get Kendall's tau and a p-value, with significance called at 0.05.
- **Seasonal decomposition** splits the series additively into trend, seasonal and residual components, then regresses the deseasonalised trend and reports the same three statistics.

The gallery below breaks mean diurnal cycles down by year, season or month, so you can see how the daily pattern shifts over the record.

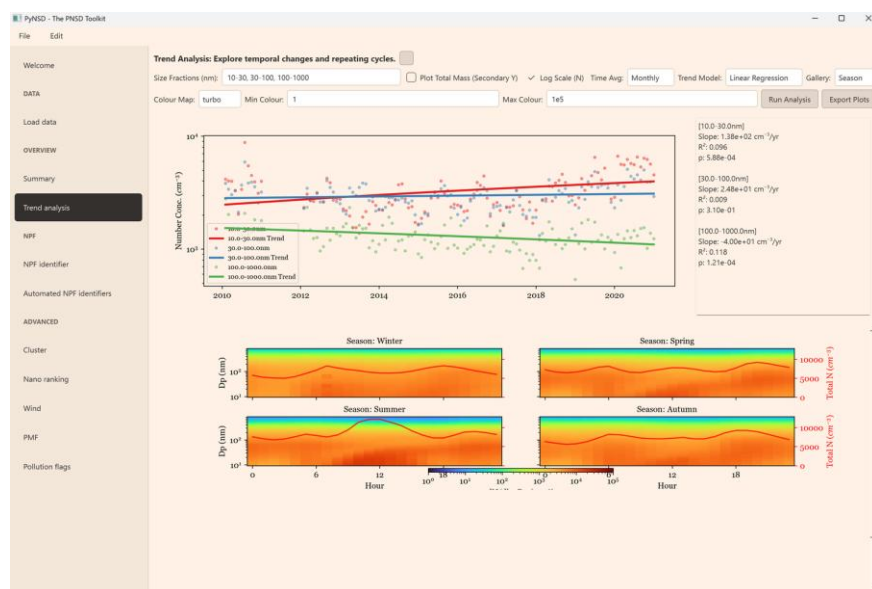


Figure 3. The *Trend analysis* panel. Three size fractions under linear regression, with slope, R^2 and p reported for each on the right, and the seasonal gallery below.

What it needs: enough record to contain several cycles of whatever period you choose. **What it will not do:** decomposition interpolates across gaps before it runs, so a record with long gaps will produce a trend component that looks smoother than the measurements justify.

7. NPF identifier

This is built to classify NPF events day-by-day manually.

Step through days with **Previous** and **Next**. Each day is drawn as a contour plot, optionally with the following day beside it in **48h Mode**. Classify it as **NPF**, **Non-NPF**, **Undefined**, **Burst**, or a label of your own.

Class	Meaning
NPF	A clear, sustained regional event. Needs a growth-rate fit.
Non-NPF	No formation. Runs the condensation sink and skips the J and GR steps.
Undefined	Messy or broken data where an event may be present but cannot be traced.
Burst	Particles appear in the smallest bins and fail to grow. J is calculated assuming a growth rate of 1 nm h^{-1} .
Custom	Your own label.

For growth rates, use **Select points** to click the mode by hand, or let PyNSD find the dominant mode in each scan and fit a line through it. **Refine fit** lets you correct individual hours by clicking. Formation

rate J is calculated between the diameter bounds you set, defaulting to 3 to 25 nm, using the real sampling interval and a number-weighted coagulation sink. $J_{1.5}$ follows by the Kerminen-Kulmala survival equation.

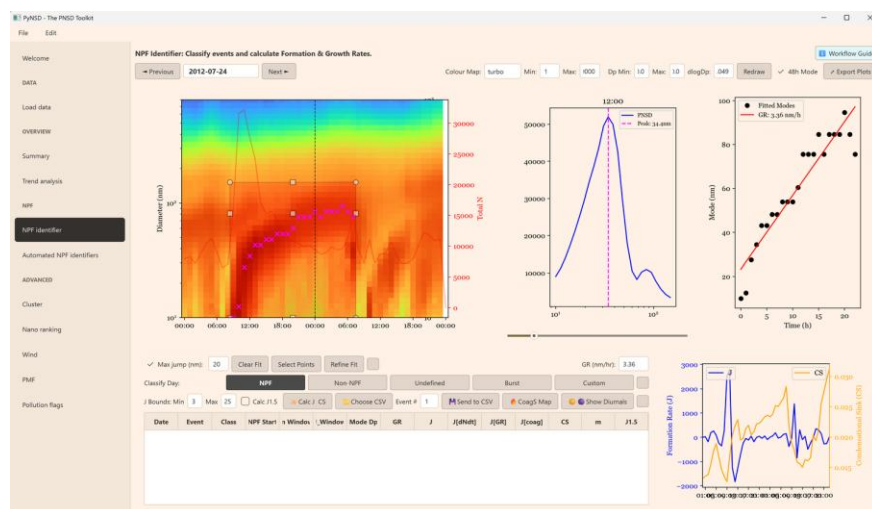


Figure 4. The **NPF identifier** on a classified event day. Crosses mark the fitted mode, the middle panel shows the spectrum at the selected hour, and the fitted growth rate and the J and CS time series appear on the right. Each classified day is appended to the table at the bottom.

What it will not do: it will not stop you fitting a growth rate to noise. The export carries every intermediate quantity so a reader can see how each number was reached.

8. Automated NPF identifiers

Two independent identifiers over the whole record, plus an automatic growth-rate tracker. Run either, or both.

The **image model** is the ResNet CNN of Kecorius et al. (2024), trained on 4,819 hand-labelled daily contour plots. PyNSD redraws each of your days in the exact style the model was trained on and returns a probability. The style matters far more than it should: the same day rendered with the wrong tick size scored 0.97 in one test and below 0.01 in another. Because the model learned from hand labels, it agrees with whatever those labellers counted as an event.

The **physical classifier** works from the measurements and follows Dal Maso et al. (2005). It tests three things: a rise in the 20 to 25 nm band against a week-pooled night-time background, a rise confined to the small end rather than across the whole distribution, and a mode that grows, starting small and early. Each day is reported with the criterion it failed, so you can check the call.

Run both. Days they agree on are usually straightforward. Days they disagree on are the interesting ones: a clean growth track the model rejects is generally a day unlike its training set, and a day the model accepts but the criteria reject is often a plume that arrived already grown.

Choose **Both** in the identifier box, leave **Track growth rates** ticked, and click **Run identifiers**. A summary line reports the class counts, the CNN event count and the median tracked growth rate. Five tabs then break the run down: **Classes**, **Mean days**, **Mean distributions**, **Growth rates** and **Agreement**.



Figure 5. The *Automated NPF identifiers* panel after a run over 3,359 days. The **Growth rates** tab compares the two independent estimates; points far from the one-to-one line are the days worth checking by hand.

What you get: **Export daily table** writes one row per day carrying the rise ratio, the selectivity, both growth-rate estimates and whether they agree, the CNN probability and each identifier's call. **What it will not do:** turning the growth tracker off makes a run much faster, but then nothing tests whether the mode grew and no day can reach class I.

The internal workings of this are quite fiddly, so email me if you want to discuss it!

9. Cluster

Group your spectra into clusters. Four steps: compare algorithms, tune the chosen one, apply it, export. See (Beddows et al., 2009; Brean et al., 2025; Dall'Osto et al., 2017) for applications.

Daily mode flattens all 24 hours into one row, so it groups whole days by their diurnal evolution. **Hourly mode** treats every hour independently, so it groups aerosol states wherever they occur. Eight algorithms are available: K-means, mini-batch K-means, agglomerative with Ward, average or complete linkage, Gaussian mixture, spectral and DBSCAN. **Compare** runs several at once and scores them by silhouette, Calinski-Harabasz and Davies-Bouldin.

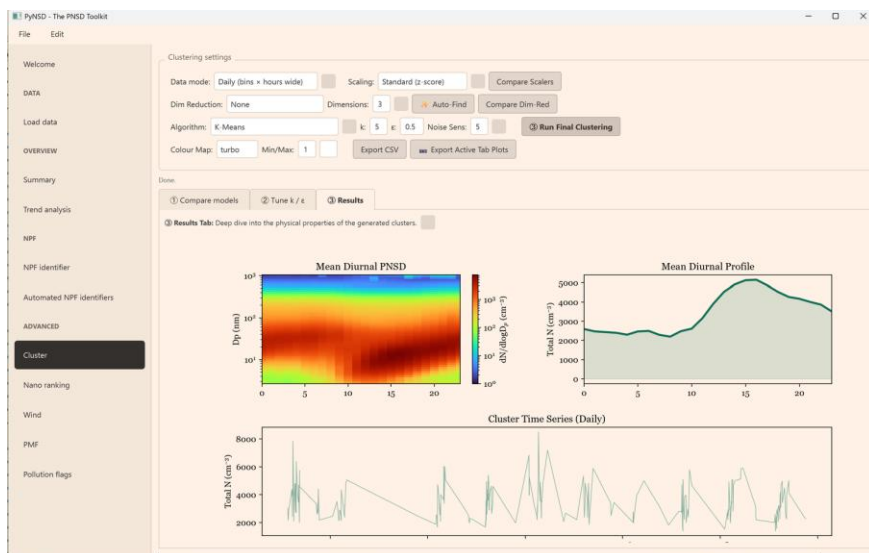


Figure 6. The **Cluster** panel on the **Results** tab, showing one cluster's mean diurnal contour, its mean diurnal total number, and when it occurs across the record. The first two tabs compare algorithms and tune k or ϵ .

What it needs: at least 20 valid hours in a day for daily mode. **What it will not do:** it will not tell you the right number of clusters. The elbow and silhouette plots narrow it down and you decide.

10. Nano ranking

Ranks days by how many particles appear in a size band you choose relative to a nighttime period, following Aliaga et al. (2023). The default band is 2.5 to 5.0 nm, with a night-time background from 21:00 to 06:00 and an active window from 06:00 to 18:00.

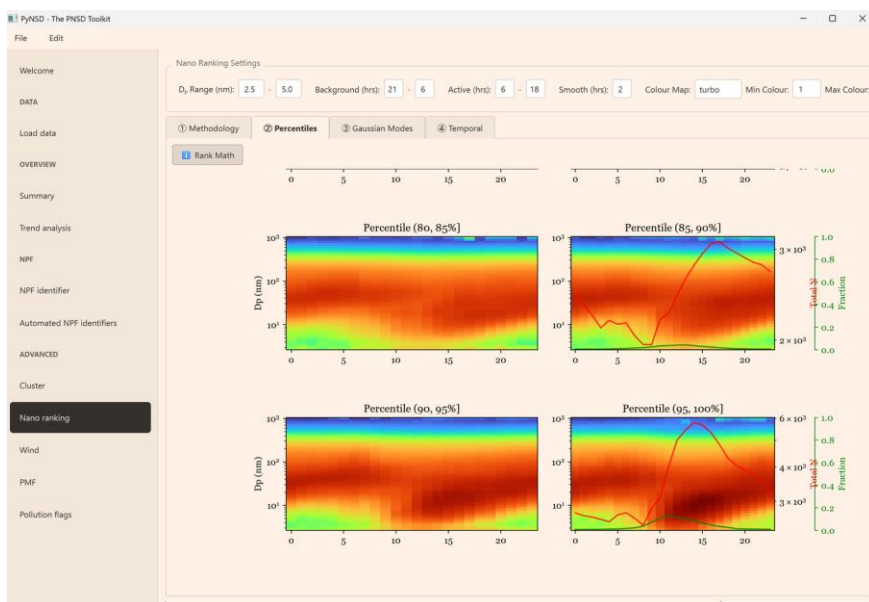


Figure 7. The **Nano ranking** panel on the **Percentiles** tab. Each panel is the mean day for a percentile band of the ranking, so the strongest days sit in the bottom right.

What you get: four tabs covering the methodology, percentile ranking, Gaussian mode fits and temporal behaviour, plus a CSV. **What it needs:** bins small enough to cover your chosen range. An instrument starting at 10 nm cannot rank on a 2.5 to 5 nm band.

11. Wind

Splits the distribution by wind direction. Load a met file with wind speed and direction, and PyNSD parses it with the same date handling as the main loader.

Wind is vector-averaged when it is resampled, so a north wind either side of an hour does not average to a south wind. Define sectors as degree ranges, defaulting to eight equal sectors. You get a rose, sector-specific mean distributions and diurnals, and both plots and data export.

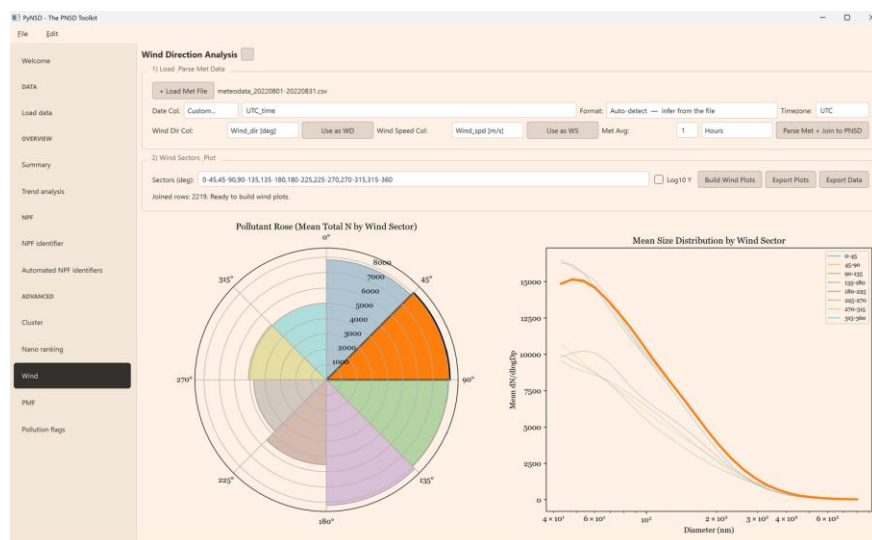


Figure 8. The *Wind* panel. Name the direction and speed columns, join to the PNSD, then build the rose and the sector distributions. Hovering a sector on the rose highlights its distribution on the right.

What it will not do: this is a conditional average, not a source apportionment. It shows what arrives from each direction and says nothing about what emitted it.

12. PMF

A complete front-end to PMF2. PyNSD writes the input matrices and the .ini file, runs the executable in batches across factor numbers and FPEAK values, then reads the output back for exploration.

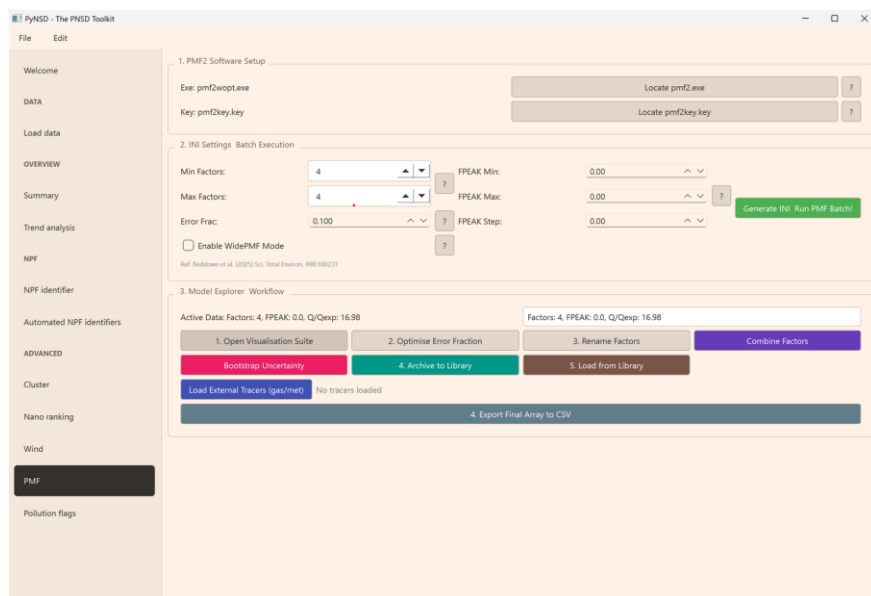


Figure 9. The PMF panel. Point it at `pmf2.exe` and your key, set the factor and FPEAK ranges, then **Generate INI Run PMF Batch**. The workflow row below runs the explorer, optimises the error fraction, renames or combines factors, and archives a solution you want to keep.

The explorer covers size and mass distributions, time series, seasonal and day-of-week composites, diurnal cycles, residuals, Q/Qexp scree plots, factor summaries, G-space scatter, tracer correlation, polar and CBPF plots and a nucleation split. **Load external tracers** brings in gas or met time series and correlates them against each factor. **Enable WidePMF mode** runs the 24-hour formulation of (Beddows et al., 2025), which treats contiguous hourly bins as the rows of the F matrix.

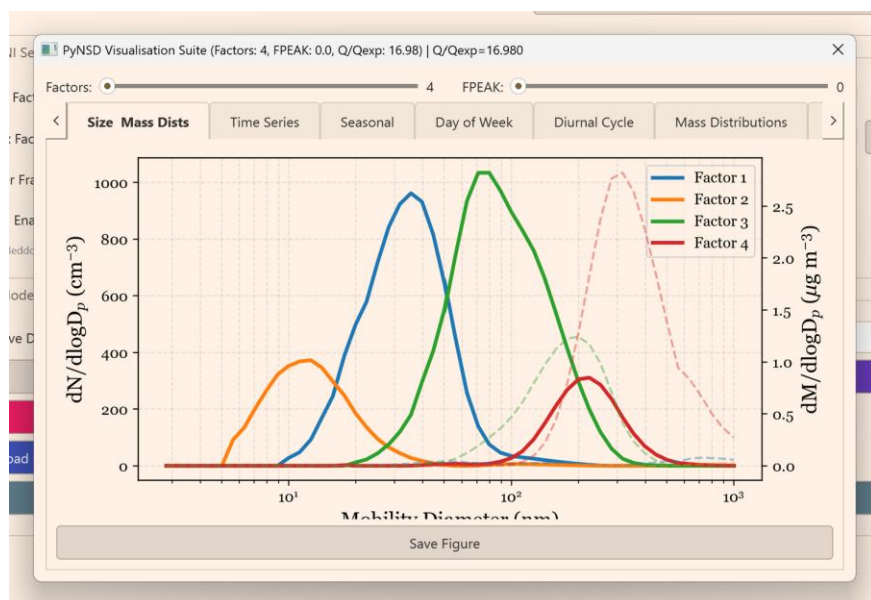


Figure 10. The visualisation suite, opened on a four-factor solution. Q/Qexp for the active run is shown in the title bar, and every tab exports independently.

What it needs: `pmf2.exe`, or `pmf2wopt.exe` for error-coefficient optimisation, and your own `pmf2key.key`. None of these are distributed with PyNSD. **What you get:** three exports. `_F_Profiles.csv` holds the factor size distributions, `_G_Raw.csv` the dimensionless time series, and `_G_Scaled_ParticleNumber.csv` the same scaled to particle number in cm^{-3} .

13. Pollution flags

Mark periods you want out of the analysis, for example, when using cruise data. Load an auxiliary file with up to two pollutant time series (e.g. BC, Nox), join it, then step through days and click any point on the contour plot to toggle its flag. The panel on the right compares the mean distribution of all rows against the unflagged ones, so you can see what the flagging is doing.

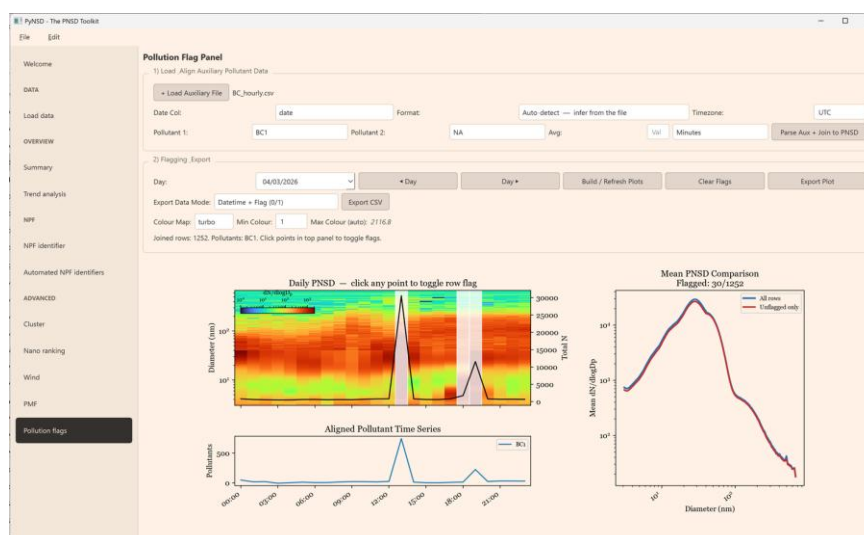


Figure 11. The *Pollution flags* panel with hourly black carbon loaded as the auxiliary series. Flagged periods are washed out in the contour plot, and the comparison on the right shows the effect of removing them.

What you get: three export modes. A timestamp and flag column, your PNSD with a flag column added, or your PNSD with the flagged rows removed.

14. What each panel exports

Panel	Exports
Load data	The corrected dataset as CSV, at any stage.
Summary, Trend, Wind, Pollution flags	Plots as PNG, PDF or SVG, at a size you set in the export dialogue.
NPF identifier	One row per classified day with GR, J, its three components, CS, m and $J_{1.5}$.
Automated NPF identifiers	<code>npf_identifications.csv</code> , one row per day, plus the summary plots.
Cluster	The cluster assignment time series as CSV, plus every plot.
Nano ranking	The ranking as CSV, plus four plots.
PMF	The F and G matrices, factor summary, nucleation sources, and every diagnostic plot.
Wind	Sector products as CSV, plus the rose and distribution figures.

Appendix A. Error messages

Message	What to do
<i>No column in this file parses as a date</i>	Set the date column by name in Load data .
<i>Date column 'x' not found</i>	Check the spelling against the column list in the message.
<i>No timestamps could be parsed</i>	The column is right and the format is wrong. Try Custom....
<i>No size-bin columns found</i>	Your headers carry no diameters. See section 3.3.
<i>Dataframe became empty</i>	Missing-data handling removed every row. Switch from Drop rows to Interpolate .
<i>Only n of N rows survived loading</i>	A warning, not an error. Open the file notes to see what went and why.
<i>Mixed diameter units in the header</i>	Some headers say nm and some say μm . Fix the file.
<i>Day/month order is ambiguous</i>	Set the date format explicitly. Do not leave this to chance.

References

- Aliaga, D., Tuovinen, S., Zhang, T., Lampilahti, J., Li, X., Ahonen, L., Kokkonen, T., Nieminen, T., Hakala, S., Paasonen, P., Bianchi, F., Worsnop, D., Kerminen, V. M., and Kulmala, M.: Nanoparticle ranking analysis: determining new particle formation (NPF) event occurrence and intensity based on the concentration spectrum of formed (sub-5 nm) particles, *Aerosol Research*, 1, 81–92, 10.5194/ar-1-81-2023, 2023.
- Beddows, D. C. S., Dall'Osto, M., and Harrison, R. M.: Cluster Analysis of Rural, Urban, and Curbside Atmospheric Particle Size Data, *Environmental Science & Technology*, 43, 4694–4700, 10.1021/es803121t, 2009.
- Beddows, D. C. S., Brean, J., Rowell, A., Merkel, M., Weinhold, K., Dall'Osto, M., and Harrison, R. M.: Wide-Positive Matrix Factorisation of particle number size distributions: A new approach accounting for cyclically changing source profiles, *Science of The Total Environment*, 998, 180231, <https://doi.org/10.1016/j.scitotenv.2025.180231>, 2025.
- Brean, J., Beddows, D. C. S., Asmi, E., Virkkula, A., Quéléver, L. L. J., Sipilä, M., Van Den Heuvel, F., Lachlan-Cope, T., Jones, A., Frey, M., Lupi, A., Park, J., Yoon, Y. J., Weller, R., Marinovich, G. L., Mulena, G. C., Harrison, R. M., and Dall'Osto, M.: Multiple eco-regions contribute to the seasonal cycle of Antarctic aerosol size distributions, *Atmos. Chem. Phys.*, 25, 1145–1162, 10.5194/acp-25-1145-2025, 2025.
- Dall'Osto, M., Beddows, D. C. S., Tunved, P., Krejci, R., Strom, J., Hansson, H. C., Yoon, Y. J., Park, K. T., Becagli, S., Udisti, R., Onasch, T., O'Dowd, C. D., Simo, R., and Harrison, R. M.: Arctic sea ice melt leads to atmospheric new particle formation, *Scientific Reports*, 7, 10, 10.1038/s41598-017-03328-1, 2017.